Lecture 9

Models of Career Concerns

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Let's Tool up: Normal Learning

Suppose I believe $\theta \sim \mathcal{N}(m, \sigma_\theta^2)$

Now I observe a signal $s = \theta + \epsilon$ with $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$

Bayesian updating on the true value of θ now takes a particularly convenient form. My posterior beliefs are that θ is distributed normally with mean

$$\overline{m} = \lambda s + (1 - \lambda)m$$

variance

$$\overline{\sigma}^2 = \lambda \sigma_\epsilon^2$$

with

$$\lambda = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\epsilon^2}$$

Career Concerns

Perhaps incentives can be created not just by contracting on performance, but based on agent's concerns about future

Principals might learn about future performance from past performance

This might make it easier to create good incentives because agent wants to appear high ability

A Simple Model

Two periods: $t \in \{1, 2\}$

Performance in each period, t, is a function of ability θ, effort e_t, and random noise ϵ_t

$$\pi_t = \theta + e_t + \epsilon_t$$

$\theta \sim \mathcal{N}(m, \sigma_\theta^2)$

$\epsilon_t \sim \mathcal{N}(0, \sigma_\epsilon^2)$

In each period agent gets $w - c(e)$

Principal chooses wage to maximize expected performance

Information

Uncertainty about θ is symmetric between principal and agent (all share same prior)

Agent and principal observe π_t each period

Only the agent observes e_t, so there is moral hazard

Principal will form beliefs about e_t (correct in equilibrium) and then try to infer θ from performance

The Game

Period 1

- Pay wage w_1
- Choose effort e_1
- Output is realized and observed: $\pi_1 = \theta + e_1 + \epsilon_1$

Period 2 (Suppose that principal believes $e_1 = \hat{e_1}$)

- Pay wage $w_2(\pi_1; \hat{e}_1)$
- Choose effort e_2
- Output realized: $\pi_2 = \theta + e_2 + \epsilon_2$

Second Period

Assume there is competition (agent can be hired elsewhere), so in second period agent will be paid expected output

In second period, agent chooses $e_2 = 0$ for certain

Second period wage is:

$$\begin{aligned} w_2(\pi_1; \hat{e}_1) &= \mathbb{E}[\pi_2 | \pi_1, \hat{e}_1] \\ &= \mathbb{E}[\theta | \pi_1, \hat{e}_1] \end{aligned}$$

Beliefs

From the principal's perspective, the quantity $\pi_1 - \hat{e}_1 = \theta + \epsilon_1$ is a noisy signal of θ

By Normal Learning, principal's posterior about θ is distributed normally with mean

$$\overline{m} = \lambda(\pi_1 - \hat{e}_1) + (1 - \lambda)m$$

with

$$\lambda = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\epsilon^2} = \left(1 + \frac{\sigma_\epsilon^2}{\sigma_\theta^2}\right)^{-1}$$

Second Period Wage

Agent when choosing e_1 knows that

$$w_2(\pi_1|\hat{e}_1) = \lambda(\pi_1 - \hat{e}_1) + (1-\lambda)m$$
$$= \lambda(\theta + e_1 + \epsilon_1 - \hat{e}_1) + (1-\lambda)m$$

All else equal, wage increases in first period effort, which increases principal's belief about θ

In any Nash equilibrium $\hat{e}_1 = e_1$ (conjectures about other player's strategy are correct), so principal isn't fooled on equilibrium path

However, in equilibrium reward must be such that agent's optimal effort is indeed $\hat{e}_1$

First Period Effort

The agent solves

$$\max_{e_1}(w_1 - c(e_1)) + \delta\left(\mathbb{E}_{\pi_1}\left[w_2(\pi_1|\hat{e}_1)|e_1\right] - c(0)\right)$$

Assuming $c(0) = 0$ and substituting $w_2(\pi_1|\hat{e}_1)$, we have

$$\max_{e_1}(w_1 - c(e_1)) + \delta\left(\mathbb{E}_{\theta+\epsilon_1}\left[\lambda(\theta + e_1 + \epsilon_1 - \hat{e}_1) + (1 - \lambda)\right.\right.$$

$$\max_{e_1}(w_1 - c(e_1)) + \delta\left(m + \lambda(e_1 - \hat{e}_1)\right)$$

This yields the following FOC

$$c'(e_1^*) = \delta\lambda$$

First Period Wage

In equilibrium $\hat{e}_1 = e_1^*$

First period wage is

$$w_1 = \mathbb{E}[\pi_1 | \textbf{prior}] = m + e_1^*$$

Interpretation

Agent chooses inefficiently low effort ($\delta\lambda < 1$)

- Benefit of effort only in next period, so discounted
- Only rewarded for the part of effort credited to ability

Agent doesn't fool principal on path, but effort is incentivized without complicated incentive contracts

- Suppose principal expected zero effort, agent would choose positive effort to fool principal and increase w_2

Comparative statics on e_1^*

- Increasing in δ
- Decreasing in noise/signal ratio: $\frac{\sigma_\epsilon^2}{\sigma_\theta^2}$

Career Concerns Model of Elections

Two periods: $t \in \{1, 2\}$

Performance in each period, t, is a function of ability of incumbent θ_I, effort e_t, and random noise ϵ_t

$$\pi_1 = \theta_I + e_1 + \epsilon_1$$

Ability for all politicians is distributed according to $\mathcal{N}(m, \sigma^2_\theta)$

$\epsilon_t \sim \mathcal{N}(0, \sigma^2_\epsilon)$

Information

Uncertainty about θ is symmetric between voter and incumbent

Voter and incumbent observe π_t each period

Only the incumbent observes e_t, so there is moral hazard

Voter has conjecture about e_t (correct in equilibrium) and then tries to learn about θ

The Game

Period 1

- Incumbent chooses effort e_1
- Output is realized and observed: $\pi_1 = \theta_I + e_1 + \epsilon_1$
- Voter can reelect politician or replace with a challenger randomly drawn from the population

Period 2

- New incumbent chooses effort e_2
- Output realized: $\pi_2 = \theta_2 + e_2 + \epsilon_2$, with θ_2 the ability of the second period incumbent

Payoffs

I gets benefit B for reelection and suffers costs $c(e_t)$ for effort

Voter cares about performance (but cannot commit to retention rule): $(\pi_1+)\pi_2$

Key differences:

- “principal” constrained by retention rule
- “principal” does not directly ‘pay’ for reward

Second Period

In second period, incumbent chooses $e_2 = 0$ for certain

Second period output is $\theta_2 + \epsilon_2$

If reelect incumbent, then second period expected output is $\mathbb{E}[\theta_I + \epsilon_2 | \pi_1, \hat{e}_1] = \mathbb{E}[\theta_I | \pi_1, \hat{e}_1]$

If replace incumbent, then second period expected output is $\mathbb{E}[\theta_C + \epsilon_2 | \pi_1, \hat{e}_1] = m$

Beliefs

The principal believes that $\pi_1 - \hat{e}_1$ is distributed normally with mean θ_I

By Normal Learning, principal's posteriors are that θ_I is distributed normally with mean

$$\overline{m} = \lambda(\pi_1 - \hat{e}_1) + (1 - \lambda)m$$

with

$$\lambda = \frac{\sigma_\theta^2}{\sigma_\theta^2 + \sigma_\epsilon^2}$$

Voting Rule

Reelect if

$$\overline{m} \geq m \iff$$

$$\lambda(\pi_1 - \hat{e}_1) + (1 - \lambda)m \geq m \iff$$

$$\pi_1 \geq m + \hat{e}_1$$

First Period Effort

The incumbent anticipates this voting behavior and takes beliefs $\hat{e}_1$ as given when choosing the optimal level of effort:

$$\max_{e_1} \Pr\left(\text{Reelect}|e_1\right) B - c(e_1)$$

The incumbent is reelected if and only if:

$$\pi_1 \geq m + \hat{e}_1 \iff$$
$$\theta_I + e_1 + \epsilon_1 \geq m + \hat{e}_1 \iff$$
$$\theta_I + \epsilon_1 \geq m + \hat{e}_1 - e_1$$

From the incumbent's perspective, LHS is a normal random variable with mean m and variance $\sigma^2 = \sigma_\theta^2 + \sigma_\epsilon^2$, so

$$\Pr\left(\text{Reelect}|e_1\right) = 1 - \Phi\left(\frac{m + \hat{e_1} - e_1 - m}{\sigma}\right) = 1 - \Phi\left(\frac{\hat{e_1} - e_1}{\sigma}\right)$$

First Period Effort (cont'd)

The incumbent solves

$$\max_{e_1} \left(1 - \Phi\left(\frac{\hat{e}_1 - e_1}{\sigma}\right)\right) B - c(e_1)$$

FOC is

$$\phi\left(\frac{\hat{e}_1 - e_1^*}{\sigma}\right)\frac{B}{\sigma} = c'(e_1^*)$$

In equilibrium, $\hat{e}_1 = e_1^*$

$$\phi(0)\frac{B}{\sigma} = c'(e_1^*)$$

Comparative Statics

$$\phi(0)\frac{B}{\sigma} = c'(e_1^*)$$

$$\Pr(Reelect) = Pr(\pi_1 \geq m + e_1^*)$$

Reelection probability increases in performance

- Seems obvious but see Wolfers' *Are Voters Rational? Evidence from Gubernatorial Elections*

Effort increasing in B

- Term limits may decrease B
- Salary and other perks of office increase B

Effort decreasing in $\sigma = \sqrt{\sigma_\theta^2 + \sigma_\epsilon^2}$

- News coverage reduces σ
- Uncertainty about pool of candidates increases σ

Incentives vs. Selection

Expected first-period performance is

$$m + e_1^*$$

Expected second-period performance, conditional on reelection, is

$$\mathbb{E}[\theta_I | \pi_1 \geq m + e_1^*] > m$$

- First-period politicians work harder (incentive effect)
- Reelected politicians have higher expected ability (competence effect)

Besley and Case

Compare performance of term-limited and non-term limited governors

Term limited governors have worse incentives for effort ($B = 0$)

Find that taxing and spending are higher under term limited

Find that effect of term limits diminishes over time (a puzzle)

Alt, Bueno de Mesquita, and Rose

Disentangle incentive and competence effects using variation in term limits

Incentive Effect: Compare 1st term eligible to 1st term ineligible

Competence Effect: Compare 1st term ineligible to 2nd term ineligible

This will also resolve the Besley-Case puzzle

Disentangling Two Effects

TABLE 4 One-Term Limits vs. Two-Term Limits

Dependent variables Expected signs on coefficients:	Log of per capita spending –		Log of per capita taxes –		Borrowing cost –		Economic growth –	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
First-term eligible (Accountability)	−0.048**	−0.065**	−0.039**	−0.039**	−5.81**	−14.04**	0.66**	0.82**
	(0.012)	(0.015)	(0.014)	(0.018)	(2.18)	(3.45)	(0.27)	(0.33)
Second-term lame duck (Competence)	−0.041**	−0.050**	−0.030**	−0.029**	−6.75**	−14.54**	0.45**	0.54*
	(0.012)	(0.015)	(0.015)	(0.018)	(2.47)	(3.44)	(0.29)	(0.32)
Sample includes governors in office at time of two-term limit adoption?	Yes	No	Yes	No	Yes	No	Yes	No
Observations	686	622	686	622	286	261	686	622
R^2	0.98	0.98	0.98	0.98	0.72	0.75	0.69	0.68

Note: The omitted category is first-term lame ducks. Controls: state income, population, percent elderly and school-aged, Democratic Governor, Democratic House, Democratic Senate, divided government, political competition in the House and Senate, governor's years of prior political experience, state-specific time trends, state fixed effects, and year fixed effects.
Robust standard errors in parentheses.
*Significant at 10% level.
**Significant at 5% level.

Resolving Besley-Case Puzzle

Incentive and Competence effects are of similar magnitudes in these data

Over time, term limits went from being one-term to two-terms

Mid-century: Term limited vs. not term limited was 1st term of both, so only effect was incentive effect

Late-Century: Term limited vs. Not term limited was 2nd term limited vs. 1st term not limited, so differed both in incentives (tends to make term limited worse) and expected ability (tends to make term limited better, here) and the effects off-set

You can only see this if you disentangle the two effects

de Janvry, Finan, and Sadoulet

Brazilian conditional cash transfer program meant to keep children in school

Large variation in success of program across cities (mean reduction in dropouts is 8%)

Cities with non-term limited mayors have, on average, a 10% reduction in dropouts

Weak evidence that reelection eligible mayors are more likely to be reelected if program is more successful

Snyder and Strömberg

Congruence of congressional district and media market as source of exogenous variation in voter information

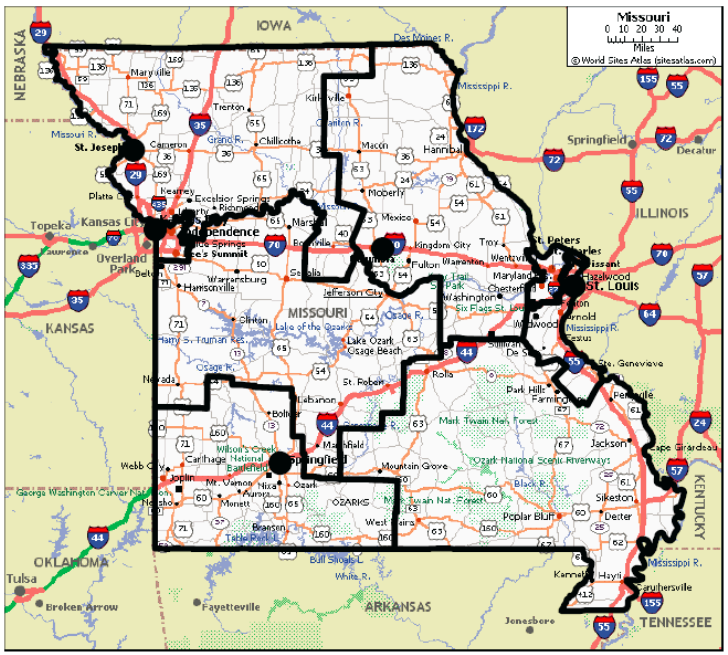
See whether more information improves performance (as suggested in the baseline model)

Congruence is high if the primary newspaper sources in a county cover primarily that county's congressional representative

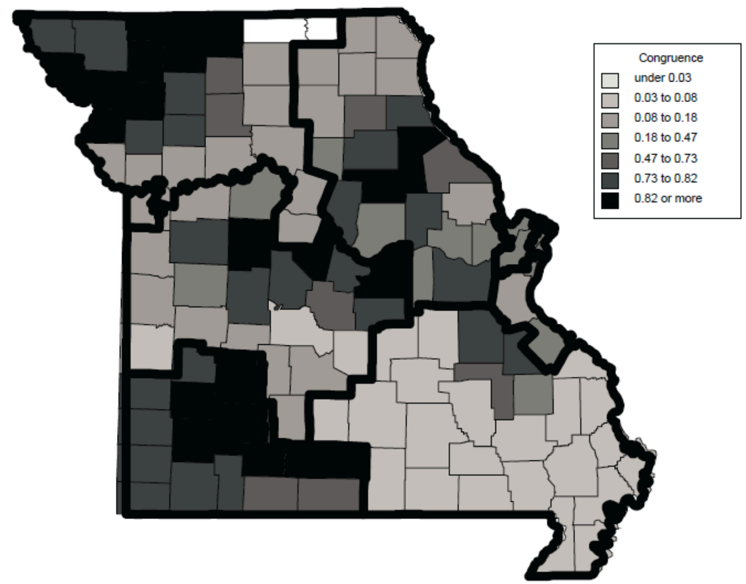
- Imagine a county near a city in the same congressional district: congruence is high
- Imagine a county near a city in a different congressional district: congruence is low

Congruence

Congressional districts



Congruence between newspaper markets and congressional districts



Identification Strategies

Comparing counties within a given state in a given year

Compare counties within a particular congressional race

Compare a particular county, that got redistricted, to itself

The Results

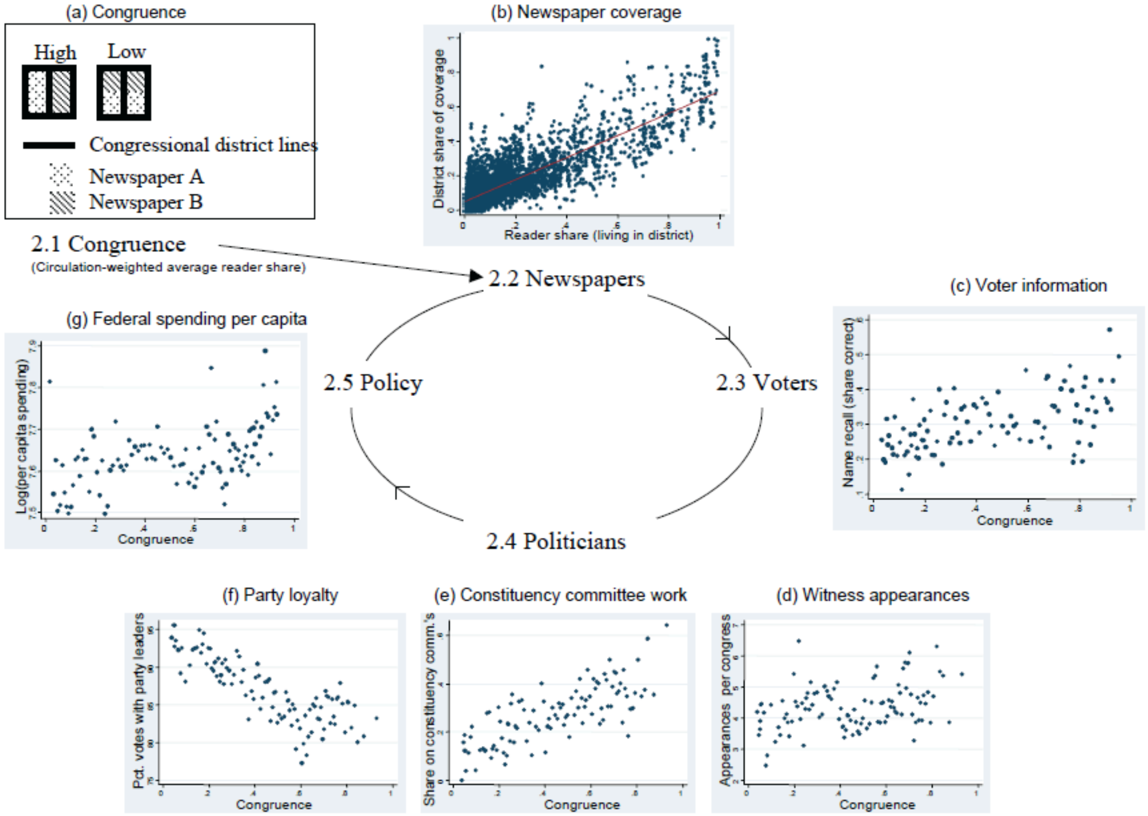


Figure 1: Structure of empirical investigation

Ferraz and Finan

Brazilian government audits cities for corruption, some before and some after election, and announces results

Let m be the mean corruption in a city audited after the election

Mayors of cities that were audited pre-election and had corruption lower than m were more likely to be reelected than mayors in un-audited cities

Mayors of cities that were audited pre-election and had corruption higher than m were less likely to be reelected than mayors in un-audited cities

No chance for behavioral response, so we only see electoral response, not change in corruption

Bias toward Incumbent

Suppose voter likes incumbent better than challenger, all else equal

- Name recognition
- Higher prior expected ability (can be derived endogenously)
- Shared partisanship

Call the bias in favor of the incumbent v

Voting Rule with Bias

Reelect if

$$\overline{m} + v \geq m \iff$$

$$\lambda(\pi_1 - \hat{e}_1) + (1 - \lambda)m \geq m - v \iff$$

$$\pi_1 \geq m + \hat{e}_1 - \frac{v}{\lambda}$$

Incumbent's Problem with Bias

The incumbent solves

$$\max_{e_1} \Pr\left(\text{Reelect}|e_1\right) B - c(e_1)$$

$$\max_{e_1} \left(1 - \Phi\left(\frac{\hat{e}_1 - e_1 - \frac{v}{\lambda}}{\sigma}\right)\right) B - c(e_1)$$

First-order condition:

$$\phi\left(\frac{\hat{e}_1 - e_1^* - \frac{v}{\lambda}}{\sigma}\right)\frac{B}{\sigma} = c'(e_1^*)$$

In equilibrium, $\hat{e}_1 = e_1^*$

$$\phi\left(\frac{-v}{\lambda\sigma}\right)\frac{B}{\sigma} = c'(e_1^*)$$

Interpretation

$$\phi\left(\frac{-v}{\lambda\sigma}\right)\frac{B}{\sigma} = c'(e_1^*)$$

Effort maximized when $v = 0$

- Entrenched incumbents reduce effort

Probability of reelection strictly increasing in v

Now the effect of increased information is complicated

- Positive if and only if v sufficiently small because noise hurts really entrenched incumbents by increasing tail risk
- Surprisingly, this implies, increased randomness in the incentive scheme can sometimes increase effort in the electoral context, unlike the contracting context

Set-Up

An agent devotes effort to two tasks (e_1, e_2)

Output on task i is $\pi_i = e_i + \epsilon_i$

The ϵs are independent and distributed $\mathcal{N}(0, \sigma_i^2)$

Some Assumptions

Mean-variance utility

- If the agent receives wages w that are the realization of a random variable W, then the agent's expected utility for these wages is $\mathbb{E}[W] - \mathbf{Var}[W]$

Quadratic costs with "complementarity" in the costs (increasing one effort increases the cost of the other effort)

$$c(e_1, e_2) = \frac{1}{2}\left(c_1 e_1^2 + c_2 e_2^2\right) + \gamma e_1 e_2$$

with $0 < \gamma < \sqrt{c_1 c_2}$

The principal only offers linear contracts:

$$w = t + w_1 \pi_1 + w_2 \pi_2$$

The Agent's Problem

Given a contract, (t, w_1, w_2) the agent solves:

$$\max_{e_1, e_2} t + w_1 e_1 + w_2 e_2 - w_1^2 \sigma_1^2 - w_2^2 \sigma_2^2 - \frac{1}{2}\left(c_1 e_1^2 + c_2 e_2^2\right) -$$

At an interior solution, we have that for each task i we need

$$w_i - c_i e_i^* - \gamma e_{-i} = 0$$

Substituting we get the following IC constraints

$$e_1^* = \frac{w_1 c_2 - \gamma w_2}{c_1 c_2 - \gamma^2}$$

$$e_2^* = \frac{w_2 c_1 - \gamma w_1}{c_1 c_2 - \gamma^2}$$

The Principal's Problem

$$\max_{t, w_1, w_2} e_1(1 - w_1) + e_2(1 - w_2) - t$$

Subject to two IC constraints:

$$e_1 = \frac{w_1 c_2 - \gamma w_2}{c_1 c_2 - \gamma^2}$$

$$e_2 = \frac{w_2 c_1 - \gamma w_1}{c_1 c_2 - \gamma^2}$$

and one IR constraint

$$t + w_1 e_1 + w_2 e_2 - w_1^2 \sigma_1^2 - w_2^2 \sigma_2^2 - \frac{1}{2}\left(c_1 e_1^2 + c_2 e_2^2\right) - \gamma e_1 e_2$$

$$t \geq \overline{u} - w_1 e_1 - w_2 e_2 + w_1^2 \sigma_1^2 + w_2^2 \sigma_2^2 + \frac{1}{2}\left(c_1 e_1^2 + c_2 e_2^2\right) +$$

Solving for the Optimal Linear Contract

Following our logic from last time, the IR constraint will bind

Substituting the IR constraint we get

$$\max_{w_1, w_2} e_1 + e_2 - \overline{u} - w_1^2\sigma_1^2 - w_2^2\sigma_2^2 - \frac{1}{2}\left(c_1 e_1^2 + c_2 e_2^2\right) - \gamma e_1 e_2$$

$$e_1 = \frac{w_1 c_2 - \gamma w_2}{c_1 c_2 - \gamma^2}$$

$$e_2 = \frac{w_2 c_1 - \gamma w_1}{c_1 c_2 - \gamma^2}$$

Solving for the Optimal Linear Contract (cont'd)

Substituting the IC constraints we get an unconstrained problem

$$\max_{w_1,w_2} \frac{w_1c_2 - \gamma w_2 + w_2c_1 - \gamma w_1}{c_1c_2 - \gamma^2} - \overline{u} - w_1^2\sigma_1^2 - w_2^2\sigma_2^2$$
$$-\frac{1}{2}c_1\left(\frac{w_1c_2 - \gamma w_2}{c_1c_2 - \gamma^2}\right)^2 - \frac{1}{2}c_2\left(\frac{w_2c_1 - \gamma w_1}{c_1c_2 - \gamma^2}\right)^2$$
$$-\gamma\left(\frac{w_1c_2 - \gamma w_2}{c_1c_2 - \gamma^2}\right)\left(\frac{w_2c_1 - \gamma w_1}{c_1c_2 - \gamma^2}\right)$$

Solving for the Optimal Linear Contract (cont'd)

Taking horrible FOCs we get:

$$w_1 = \frac{c_2 - \gamma + \gamma w_2}{2\sigma_1^2(c_1 c_2 - \gamma^2) + c_2}$$
$$w_2 = \frac{c_1 - \gamma + \gamma w_1}{2\sigma_2^2(c_1 c_2 - \gamma^2) + c_1}$$

Substituting, we have:

$$w_1^* = \frac{1 + 2(c_2 - \gamma)\sigma_2^2}{1 + 2c_1\sigma_1^2 + 2c_2\sigma_2^2 + 4\sigma_1^2\sigma_2^2(c_1 c_2 - \gamma^2)}$$
$$w_2^* = \frac{1 + 2(c_1 - \gamma)\sigma_1^2}{1 + 2c_1\sigma_1^2 + 2c_2\sigma_2^2 + 4\sigma_1^2\sigma_2^2(c_1 c_2 - \gamma^2)}$$

Interpretation

$$w_1^* = \frac{1 + 2(c_2 - \gamma)\sigma_2^2}{1 + 2c_1\sigma_1^2 + 2c_2\sigma_2^2 + 4\sigma_1^2\sigma_2^2(c_1c_2 - \gamma^2)}$$

$$w_2^* = \frac{1 + 2(c_1 - \gamma)\sigma_1^2}{1 + 2c_1\sigma_1^2 + 2c_2\sigma_2^2 + 4\sigma_1^2\sigma_2^2(c_1c_2 - \gamma^2)}$$

- As task 2 becomes harder to measure/observe (σ_2 gets bigger) both w_2^* and w_1^* get smaller
- If hard to give incentives on one task, weaken incentives on the other task to avoid effort substitution